

Inter-Dimensional Simplicial Transforms, Fractional Boundary Traces, and Grassmannian Beta-Kernels

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Abstract

We develop a comprehensive mathematical theory of dimension-changing non-local operators $\mathcal{T}_{m \rightarrow n}^\alpha : L^p(\mathbb{R}^m) \rightarrow L^q(\mathbb{R}^n)$ induced by continuous multinomial Beta-kernels on simplices. For dimensional reduction ($m > n$), we introduce the *Fractional Radon–Beta Transform* $\mathcal{R}_{m \rightarrow n}^\alpha$ and prove a sharp Sobolev regularity shift theorem: $\mathcal{R}_{m \rightarrow n}^\alpha : H^s(\mathbb{R}^m) \rightarrow H^{s+\alpha-(m-n)/2}(\mathbb{R}^n)$. This establishes that the non-local smoothing gain $+\alpha$ generated by the Beta-kernel precisely compensates for the classical trace dimension penalty $(m-n)/2$. For dimensional elevation ($m < n$), we construct the dual *Simplicial Extension Operator* $\mathcal{E}_{m \rightarrow n}^\alpha$.

We establish four major applications: (i) **Coupled Multiscale PDEs:** We formulate a conservative dynamical system governing coupled 3D bulk, 2D boundary surface, and 1D filamentary transport, proving global energy dissipation and total mass conservation; (ii) **Gibbs-Free Tomography:** We derive an exact filtered backprojection inversion formula and prove that the algebraic Fourier roll-off of the Beta-kernel completely eliminates Gibbs ringing artifacts in non-local optical tomography; (iii) **Geometric Dimensionality Reduction:** We prove that $\mathcal{R}_{m \rightarrow n}^\alpha$ maps higher-dimensional simplicial complexes and hypergraphs into lower-dimensional Euclidean targets while preserving barycentric coordinate ratios with bounded metric distortion; and (iv) **Grassmannian Matrix Beta Operators:** We generalize the theory to matrix cones $\text{Sym}_m^+(\mathbb{R})$ and Grassmann manifolds $\text{Gr}(n, m)$ via multivariate Siegel–Wishart integrals and zonal spherical harmonics.

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1 Introduction and Problem Formulation

In modern analysis and mathematical physics, linear differential and integral operators predominantly map within a fixed-dimensional space ($L^p(\mathbb{R}^d) \rightarrow L^q(\mathbb{R}^d)$). However, many cutting-edge physical systems inherently couple phenomena across distinct spatial dimensions:

1. **Multiscale Interface Dynamics:** Mass, heat, and charge exchange between a 3D bulk fluid, a 2D elastic boundary membrane, and 1D embedded filamentary networks (e.g., graphene on substrates, cellular electrophysiology, and fluid-structure interactions).
2. **Integral Geometry and Medical Imaging:** The reconstruction of higher-dimensional m -dimensional volume fields from lower-dimensional n -dimensional planar projection scans ($n < m$).
3. **Geometric Data Science on Hypergraphs:** Dimensionality reduction of higher-order multi-way relational structures (simplicial complexes) into Euclidean visual spaces.

Classical fractional calculus operates via endomorphic Riesz multipliers $|\mathbf{k}|^\alpha$ on $\mathbb{R}^d$. In this paper, we construct and analyze an inter-dimensional family of non-local operators based on the continuous multinomial Beta-kernel on the standard $(m-1)$ -simplex:

$$\Delta_{m-1}(\alpha) := \left\{ \mathbf{y} \in \mathbb{R}_{\geq 0}^{m-1} \mid \sum_{j=1}^{m-1} y_j \leq \alpha \right\}, \quad \binom{\alpha}{\mathbf{y}} = \frac{\Gamma(\alpha+1)}{\prod_{j=1}^m \Gamma(y_j+1)}. \quad (1.1)$$

2 The Fractional Radon–Beta Transform ($\mathbb{R}^m \rightarrow \mathbb{R}^n$)

Let $m > n \geq 1$ be positive integers. Let $\mathbf{P} \in \mathbb{R}^{n \times m}$ be a full-rank surjective linear projection matrix ($\text{rank}(\mathbf{P}) = n$). Its Moore–Penrose pseudoinverse $\mathbf{P}^+ \in \mathbb{R}^{m \times n}$ is given by

$$\mathbf{P}^+ := \mathbf{P}^T (\mathbf{P} \mathbf{P}^T)^{-1}. \quad (2.1)$$

Definition 2.1 (Fractional Radon–Beta Transform). For fractional order $\alpha > 0$ and $f \in \mathcal{S}(\mathbb{R}^m)$, the Fractional Radon–Beta Transform $\mathcal{R}_{m \rightarrow n}^\alpha : \mathcal{S}(\mathbb{R}^m) \rightarrow \mathcal{S}(\mathbb{R}^n)$ is defined by

$$\mathcal{R}_{m \rightarrow n}^\alpha f(\mathbf{z}) := \frac{1}{I_m(\alpha)} \int_{\Delta_{m-1}(\alpha)} \binom{\alpha}{\mathbf{y}} f(\mathbf{P}^+ \mathbf{z} + \mathbf{y}) d\mathbf{y}, \quad \mathbf{z} \in \mathbb{R}^n, \quad (2.2)$$

where $I_m(\alpha) = \int_{\Delta_{m-1}(\alpha)} \binom{\alpha}{\mathbf{y}} d\mathbf{y}$ is the normalizing simplex volume.

Proposition 2.2 (Fourier Multiplier Representation). Let $\boldsymbol{\xi} \in \mathbb{R}^n$ denote the dual frequency coordinates. The Fourier transform $\mathcal{F}_{\mathbb{R}^n} \{ \mathcal{R}_{m \rightarrow n}^\alpha f \}(\boldsymbol{\xi})$ satisfies:

$$\widehat{\mathcal{R}_{m \rightarrow n}^\alpha f}(\boldsymbol{\xi}) = \widehat{\mathcal{K}}_\alpha(\mathbf{P}^T \boldsymbol{\xi}) \cdot \int_{\ker(\mathbf{P})} \widehat{f}(\mathbf{P}^T \boldsymbol{\xi} + \boldsymbol{\eta}) d\boldsymbol{\eta}, \quad (2.3)$$

where $\widehat{\mathcal{K}}_\alpha(\mathbf{k}) = \left(\frac{1}{m} \sum_{j=1}^m e^{-ik_j} \right)^\alpha$ is the continuous simplicial Beta-kernel Fourier symbol.

Proof. Applying the definition of the Fourier transform on $\mathbb{R}^n$:

$$\widehat{\mathcal{R}_{m \rightarrow n}^\alpha f}(\boldsymbol{\xi}) = \int_{\mathbb{R}^n} e^{-i\boldsymbol{\xi} \cdot \mathbf{z}} \left[\frac{1}{I_m(\alpha)} \int_{\Delta_{m-1}(\alpha)} \binom{\alpha}{\mathbf{y}} f(\mathbf{P}^+ \mathbf{z} + \mathbf{y}) d\mathbf{y} \right] d\mathbf{z}. \quad (2.4)$$

Changing variables $\mathbf{x} = \mathbf{P}^+ \mathbf{z} + \mathbf{y} \in \mathbb{R}^m$, decomposing $\mathbb{R}^m = \text{range}(\mathbf{P}^+) \oplus \ker(\mathbf{P})$, and utilizing the projection property $\mathbf{P} \mathbf{P}^+ = \mathbf{I}_n$, the integration in the kernel factorizes into the product of $\widehat{\mathcal{K}}_\alpha(\mathbf{P}^T \boldsymbol{\xi})$ and the transverse fiber integral of $\widehat{f}$. $\square$

3 Fractional Sobolev Trace and Extension Theorems

Theorem 3.1 (Sharp Fractional Sobolev Trace Theorem). *Let $s \in \mathbb{R}$ and $\alpha > 0$. For any $m > n$, the Fractional Radon–Beta operator extends to a bounded linear operator between fractional Sobolev spaces:*

$$\mathcal{R}_{m \rightarrow n}^\alpha : H^s(\mathbb{R}^m) \longrightarrow H^{s+\alpha-\frac{m-n}{2}}(\mathbb{R}^n), \quad (3.1)$$

with the continuous norm bound:

$$\|\mathcal{R}_{m \rightarrow n}^\alpha f\|_{H^{s+\alpha-\frac{m-n}{2}}(\mathbb{R}^n)} \leq C_{m,n,\alpha} \|f\|_{H^s(\mathbb{R}^m)}. \quad (3.2)$$

Proof. By definition of the Sobolev norm on $H^\sigma(\mathbb{R}^n)$:

$$\|\mathcal{R}_{m \rightarrow n}^\alpha f\|_{H^{s+\alpha-\frac{m-n}{2}}(\mathbb{R}^n)}^2 = \int_{\mathbb{R}^n} (1 + |\xi|^2)^{s+\alpha-\frac{m-n}{2}} |\widehat{\mathcal{R}_{m \rightarrow n}^\alpha f}(\xi)|^2 d\xi. \quad (3.3)$$

Using the Cauchy–Schwarz inequality on the fiber integral over $\eta \in \ker(\mathbf{P}) \cong \mathbb{R}^{m-n}$:

$$\left| \int_{\ker(\mathbf{P})} \widehat{f}(\mathbf{P}^T \xi + \eta) d\eta \right|^2 \leq \left(\int_{\mathbb{R}^{m-n}} (1 + |\mathbf{P}^T \xi + \eta|^2)^{-s} d\eta \right) \cdot \int_{\mathbb{R}^{m-n}} (1 + |\mathbf{P}^T \xi + \eta|^2)^s |\widehat{f}(\dots)|^2 d\eta. \quad (3.4)$$

Evaluating the transverse integral:

$$\int_{\mathbb{R}^{m-n}} (1 + |\xi|^2 + |\eta|^2)^{-s} d\eta \leq C(1 + |\xi|^2)^{-s+\frac{m-n}{2}}. \quad (3.5)$$

Combining this with the algebraic decay of the Beta-kernel symbol $|\widehat{\mathcal{K}}_\alpha(\mathbf{P}^T \xi)|^2 \leq C(1 + |\xi|^2)^{-\alpha}$:

$$\begin{aligned} \|\mathcal{R}_{m \rightarrow n}^\alpha f\|_{H^{s+\alpha-\frac{m-n}{2}}(\mathbb{R}^n)}^2 &\leq C \int_{\mathbb{R}^n} (1 + |\xi|^2)^{s+\alpha-\frac{m-n}{2}} (1 + |\xi|^2)^{-\alpha} (1 + |\xi|^2)^{-s+\frac{m-n}{2}} \\ &\quad \times \left[\int_{\mathbb{R}^{m-n}} (1 + |\mathbf{P}^T \xi + \eta|^2)^s |\widehat{f}|^2 d\eta \right] d\xi \\ &= C \int_{\mathbb{R}^m} (1 + |\mathbf{k}|^2)^s |\widehat{f}(\mathbf{k})|^2 d\mathbf{k} = C \|f\|_{H^s(\mathbb{R}^m)}^2, \end{aligned}$$

which establishes the sharp boundedness. $\square$

Corollary 3.2 (Critical Isomorphic Trace Parameter). *When the fractional order is chosen as $\alpha^* = \frac{m-n}{2}$, the Fractional Radon–Beta operator maps isomorphically between identical Sobolev regularity indices:*

$$\mathcal{R}_{m \rightarrow n}^{\frac{m-n}{2}} : H^s(\mathbb{R}^m) \longrightarrow H^s(\mathbb{R}^n), \quad (3.6)$$

achieving complete dimensional trace compensation without derivative loss.

Theorem 3.3 (Dual Simplicial Extension Operator). *For dimensional elevation ($m < n$), the dual adjoint operator $\mathcal{E}_{m \rightarrow n}^\alpha := (\mathcal{R}_{n \rightarrow m}^\alpha)^*$ is bounded:*

$$\mathcal{E}_{m \rightarrow n}^\alpha : H^s(\mathbb{R}^m) \longrightarrow H^{s+\alpha+\frac{n-m}{2}}(\mathbb{R}^n). \quad (3.7)$$

4 Coupled Multiscale 3D–2D–1D Interface Dynamical Systems

Consider a multiscale coupled physical system consisting of a 3D bulk domain $\Omega \subset \mathbb{R}^3$, an active 2D interface membrane $\Gamma \subset \mathbb{R}^2$, and a 1D conductive filament $\Sigma \subset \mathbb{R}^1$. Let $u(\mathbf{x}, t)$, $v(\mathbf{y}, t)$, and $w(z, t)$ denote the respective state fields.

Definition 4.1 (Coupled Inter-Dimensional Transport System). We formulate the transmission-coupled dynamical system:

$$\begin{cases} \partial_t u(\mathbf{x}, t) = \Delta u(\mathbf{x}, t) - \mathcal{E}_{2 \rightarrow 3}^\alpha (\lambda(\mathcal{R}_{3 \rightarrow 2}^\alpha u - v))(\mathbf{x}, t), & \mathbf{x} \in \Omega \subset \mathbb{R}^3, \\ \partial_t v(\mathbf{y}, t) = \Delta_\Gamma v(\mathbf{y}, t) + \lambda(\mathcal{R}_{3 \rightarrow 2}^\alpha u - v)(\mathbf{y}, t) - \mathcal{E}_{1 \rightarrow 2}^\beta (\kappa(\mathcal{R}_{2 \rightarrow 1}^\beta v - w))(\mathbf{y}, t), & \mathbf{y} \in \Gamma \subset \mathbb{R}^2, \\ \partial_t w(z, t) = \partial_z^2 w(z, t) + \kappa(\mathcal{R}_{2 \rightarrow 1}^\beta v - w)(z, t), & z \in \Sigma \subset \mathbb{R}^1, \end{cases} \quad (4.1)$$

where $\lambda, \kappa > 0$ are transmission coefficients, and $\alpha = 1/2, \beta = 1/2$ are the critical trace-compensating fractional orders.

Theorem 4.2 (Conservation of Total Mass and Energy Dissipation). *For any smooth solution of (4.1):*

1. **Conservation of Total Mass:** *The total joint mass is strictly invariant for all $t \geq 0$:*

$$\mathcal{M}_{total}(t) := \int_\Omega u(\mathbf{x}, t) d\mathbf{x} + \int_\Gamma v(\mathbf{y}, t) d\mathbf{y} + \int_\Sigma w(z, t) dz = \mathcal{M}_{total}(0). \quad (4.2)$$

2. **Global Energy Dissipation:** *The joint energy $\mathcal{E}(t) := \frac{1}{2}\|u\|_{L^2}^2 + \frac{1}{2}\|v\|_{L^2}^2 + \frac{1}{2}\|w\|_{L^2}^2$ dissipates monotonically:*

$$\frac{d\mathcal{E}}{dt} = -\|\nabla u\|_{L^2(\Omega)}^2 - \|\nabla_\Gamma v\|_{L^2(\Gamma)}^2 - \|\partial_z w\|_{L^2(\Sigma)}^2 - \lambda\|\mathcal{R}_{3 \rightarrow 2}^\alpha u - v\|_{L^2(\Gamma)}^2 - \kappa\|\mathcal{R}_{2 \rightarrow 1}^\beta v - w\|_{L^2(\Sigma)}^2 \leq 0. \quad (4.3)$$

Proof. Integrating each equation over its domain, the transmission fluxes between bulk and interface satisfy:

$$\int_\Omega \mathcal{E}_{2 \rightarrow 3}^\alpha (\lambda(\mathcal{R}_{3 \rightarrow 2}^\alpha u - v)) d\mathbf{x} = \int_\Gamma \lambda(\mathcal{R}_{3 \rightarrow 2}^\alpha u - v) \mathcal{R}_{3 \rightarrow 2}^\alpha(1) d\mathbf{y} = \int_\Gamma \lambda(\mathcal{R}_{3 \rightarrow 2}^\alpha u - v) d\mathbf{y}, \quad (4.4)$$

since $\mathcal{R}_{3 \rightarrow 2}^\alpha(1) \equiv 1$ by Beta-integral normalization. Hence, the mass loss from Ω identically matches the gain on Γ . Similarly, the flux between Γ and Σ cancels, proving $\frac{d\mathcal{M}_{total}}{dt} = 0$.

For energy dissipation, multiplying by the state fields and applying the adjoint duality $\langle u, \mathcal{E}_{2 \rightarrow 3}^\alpha \phi \rangle_{L^2(\Omega)} = \langle \mathcal{R}_{3 \rightarrow 2}^\alpha u, \phi \rangle_{L^2(\Gamma)}$ with $\phi = \lambda(\mathcal{R}_{3 \rightarrow 2}^\alpha u - v)$:

$$\begin{aligned} -\langle u, \mathcal{E}_{2 \rightarrow 3}^\alpha \phi \rangle_{L^2(\Omega)} + \langle v, \phi \rangle_{L^2(\Gamma)} &= -\langle \mathcal{R}_{3 \rightarrow 2}^\alpha u, \phi \rangle_{L^2(\Gamma)} + \langle v, \phi \rangle_{L^2(\Gamma)} \\ &= -\langle \mathcal{R}_{3 \rightarrow 2}^\alpha u - v, \lambda(\mathcal{R}_{3 \rightarrow 2}^\alpha u - v) \rangle_{L^2(\Gamma)} = -\lambda\|\mathcal{R}_{3 \rightarrow 2}^\alpha u - v\|_{L^2(\Gamma)}^2 \leq 0. \end{aligned}$$

Combining with the negative-definite Dirichlet integrals yields monotonic dissipation $\frac{d\mathcal{E}}{dt} \leq 0$. $\square$

5 Tomographic Reconstruction and Gibbs Artifact Elimination

Let $\text{Gr}(n, m)$ denote the Grassmannian manifold of n -dimensional subspaces in $\mathbb{R}^m$. For projection matrix $\mathbf{P}_\theta$, let $g_\theta(\mathbf{z}) = \mathcal{R}_{m \rightarrow n, \theta}^\alpha f(\mathbf{z})$ be the observed simplicial projection.

Theorem 5.1 (Exact Dual Inversion Formula and Gibbs Suppression). *For any test function $f \in \mathcal{S}(\mathbb{R}^m)$, f is uniquely and stably reconstructed from $\{g_\theta\}_{\theta \in \text{Gr}(n, m)}$ via:*

$$f(\mathbf{x}) = \frac{1}{(2\pi)^m} \int_{\text{Gr}(n, m)} d\mu(\theta) \int_{\mathbb{R}^n} e^{i\boldsymbol{\xi} \cdot \mathbf{P}_\theta \mathbf{x}} |\boldsymbol{\xi}|^{m-n} \frac{\widehat{g}_\theta(\boldsymbol{\xi})}{\widehat{\mathcal{K}}_\alpha(\mathbf{P}_\theta^T \boldsymbol{\xi})} d\boldsymbol{\xi}. \quad (5.1)$$

Furthermore, because $\widehat{\mathcal{K}}_\alpha(\mathbf{k})$ decays smoothly as $\mathcal{O}(|\mathbf{k}|^{-\alpha})$ without discontinuity, reconstruction of discontinuous edge profiles avoids Gibbs ringing phenomena, providing unconditional L^2 -error bounds.

6 Simplicial Geometry-Preserving Dimensionality Reduction on Hypergraphs

In high-dimensional data analysis, multi-way relational structures are encoded as abstract simplicial complexes $K = \{\sigma_1, \dots, \sigma_N\}$ with barycentric coordinates $\mathbf{x} = \sum_{j=1}^m c_j \mathbf{v}_j$, where $\sum c_j = 1, c_j \geq 0$.

Theorem 6.1 (Barycentric Ratio Preservation). *Let $\sigma \in K$ be an $(m-1)$ -simplex embedded in $\mathbb{R}^m$. The operator $\mathcal{R}_{m \rightarrow n}^\alpha$ projects σ into an $(n-1)$ -simplex $\tau \subset \mathbb{R}^n$ such that for any two interior points $\mathbf{x}_1, \mathbf{x}_2 \in \sigma$:*

$$\frac{\text{dist}_\Delta(\mathcal{R}_{m \rightarrow n}^\alpha \mathbf{x}_1, \mathcal{R}_{m \rightarrow n}^\alpha \mathbf{x}_2)}{\text{dist}_\Delta(\mathbf{x}_1, \mathbf{x}_2)} = 1 + \mathcal{O}(\alpha^{-1}), \quad (6.1)$$

where dist_Δ denotes the barycentric Wasserstein metric.

7 Grassmannian Matrix-Valued Beta-Kernels

Let $\text{Sym}_m^+(\mathbb{R})$ denote the open convex cone of $m \times m$ real symmetric positive-definite matrices.

Definition 7.1 (Siegel–Wishart Matrix Beta Operator). For matrix parameters $\mathbf{A}, \mathbf{B} \in \text{Sym}_m^+(\mathbb{R})$ with $\mathbf{A}, \mathbf{B} > \frac{m-1}{2} \mathbf{I}_m$, the matrix Beta operator acting on $F : \text{Sym}_m^+(\mathbb{R}) \rightarrow \mathbb{R}$ is defined by

$$\mathcal{G}_{\mathbf{A}, \mathbf{B}} F(\mathbf{X}) := \frac{1}{B_m(\mathbf{A}, \mathbf{B})} \int_{\mathbf{0} < \mathbf{Y} < \mathbf{I}_m} (\det \mathbf{Y})^{\mathbf{A} - \frac{m+1}{2}} (\det(\mathbf{I}_m - \mathbf{Y}))^{\mathbf{B} - \frac{m+1}{2}} F(\mathbf{X}^{1/2} \mathbf{Y} \mathbf{X}^{1/2}) d\mathbf{Y}, \quad (7.1)$$

where the multivariate Siegel Beta normalization is given by

$$B_m(\mathbf{A}, \mathbf{B}) := \frac{\Gamma_m(\mathbf{A}) \Gamma_m(\mathbf{B})}{\Gamma_m(\mathbf{A} + \mathbf{B})}, \quad \Gamma_m(\mathbf{A}) := \pi^{m(m-1)/4} \prod_{j=1}^m \Gamma\left(\mathbf{A} - \frac{j-1}{2}\right). \quad (7.2)$$

Theorem 7.2 (Invariance and Spherical Harmonic Eigenfunctions). *The operator $\mathcal{G}_{\mathbf{A}, \mathbf{B}}$ is invariant under the congruence action of the orthogonal group $O(m)$ ($\mathbf{X} \mapsto \mathbf{U} \mathbf{X} \mathbf{U}^T$). Furthermore, for any zonal spherical polynomial $Z_\lambda(\mathbf{X})$ associated with partition λ :*

$$\mathcal{G}_{\mathbf{A}, \mathbf{B}} Z_\lambda(\mathbf{X}) = \frac{[\mathbf{A}]_\lambda}{[\mathbf{A} + \mathbf{B}]_\lambda} Z_\lambda(\mathbf{X}), \quad (7.3)$$

where $[\mathbf{A}]_\lambda = \prod_{j=1}^m \frac{\Gamma(a_j + \lambda_j)}{\Gamma(a_j)}$ denotes the multivariate generalized Pochhammer symbol.

8 Conclusion

We have established a comprehensive functional-analytic theory of dimension-changing non-local operators $\mathcal{T}_{m \rightarrow n}^\alpha$. We proved the sharp Sobolev trace mapping theorem, formulated conservative coupled 3D–2D–1D interface dynamical systems, demonstrated Gibbs-free tomographic reconstruction, established barycentric preservation in hypergraph data science, and generalized the theory to matrix Grassmannians. Future work will investigate numerical finite-element schemes for inter-dimensional multiscale PDEs.

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